

Yasah Sai Chandra Borusu

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EDUCATION

George Mason University, College of Engineering & Computing

B.S. in Computer & Information Sciences

Fairfax, VA

Aug 2022 – Dec 2026

EXPERIENCE

Software Engineer (Founding Team)

Aug 2025 – Apr 2026

TheCollegeTech

Hybrid

- Co-founded a **4-person** engineering team and shipped **3 production modules** — Learning, Campus, and Placement Management — covering attendance & gradebook, timetables, applications/shortlisting, and notifications.
- Built a reusable **React + TypeScript** component library of **25+** tables, forms, and modals with shared validation and typed API clients, cutting repeated UI work across all three modules.
- Designed **30+ Python** REST endpoints over relational **SQL** models with filtering, pagination, role-based access, parameterized queries, and schema migrations; authored unit/integration tests, set up **CI/CD** pipelines, and reviewed teammates' PRs.

Cloud & AI Intern

May 2025 – Jul 2025

Astrani Technology Solutions

Ashburn, VA (Hybrid)

- Built **AI-driven business insights** tools on **AWS Bedrock** over **S3/DynamoDB** datasets; benchmarked **OpenAI**, **DeepSeek**, and **Mistral** for CX analytics, summarization, and recommendations.
- Led **data engineering** migration from on-prem to **AWS RDS (PostgreSQL)**, achieving nearly **70%** infrastructure cost savings; designed ETL workflows and parameterized SQL for downstream reporting.
- Recommended a domain-specific **small-language-model (SLM)** strategy to cut inference cost versus large LLMs; supported image classification, text processing, and chatbot work with experiment tracking.

Undergraduate Teaching Assistant (Python)

Aug 2023 – Present

College of Engineering & Computing, George Mason University

Fairfax, VA

- Provided one-on-one debugging and conceptual support to **150+ students per semester** across labs, projects, and coursework, diagnosing root causes of errors and walking students through fixes until they could apply the concept independently.
- Served as a recurring point of contact across **6+ semesters** (Aug 2023–Present) for students stuck on Python fundamentals, project logic, and lab assignments, translating instructor material into explanations tailored to each student's specific gap in understanding.

Software Engineer Intern

May 2024 – Aug 2024

Rise Consultancy

Remote

- Shipped front-end and back-end features for client-facing web applications in **JavaScript**, **React**, and **HTML/CSS**, working within an agile engineering team across the full development lifecycle.
- Integrated **REST APIs** into responsive UI components and validated request/response contracts with **Postman**; queried and debugged **SQL** data issues and managed changes through **Git** branch and review workflows.

PROJECTS

Brevify: Banking Compliance Disclosure Pipeline — *Python, OpenAI API, Streamlit*

- Built a multi-stage pipeline that parses federal banking regulations (Reg Z, Reg DD) from the **eCFR API** into structured rule and glossary datasets via **OpenAI**-powered extraction (369 rule records, 50 glossary terms across 266 legal-unit chunks).
- Designed a two-call **LLM** claim-classification engine that evaluates marketing claims against the extracted rule set, returning risk level, required disclosures, and missing product terms; shipped a **Streamlit** frontend with single-claim and batch-CSV analysis modes.

MindEase: AI Task Planning Chatbot & Dashboard — *Next.js, TypeScript, Supabase (Postgres + Auth)*

- Built and deployed a chat app converting natural language into prioritized tasks and a daily action plan, with multi-task capture per message, subtask breakdown, and a live workload dashboard.
- Architected a **Next.js + TypeScript** frontend on **Supabase Postgres** with SSR auth, **row-level security** policies, typed database contracts, and REST routes for chat parsing, task CRUD, and summaries.

BRIK-E: Smart-Building Hazard Detection — *Python, PyTorch, YOLOv8/SegFormer, OpenCV, FastAPI*

- Trained and deployed custom computer-vision models (**two-stage** detector-to-classifier verifier) to detect spills, trash, and lost items on CCTV feeds, calibrating cameras to floor plans via **homography** and dispatching a real-time **MQTT** event pipeline for robot response and live tracking.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, PostgreSQL), Java, C/C++, TypeScript/JavaScript

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, LangChain, PyTorch, OpenCV, Pydantic, Asyncio

Cloud & Databases: GCP, AWS (Bedrock, S3, DynamoDB, RDS), Microsoft Azure (Fundamentals), Supabase

Web & Tools: REST APIs, HTML/CSS/XML, Git, Docker, Postman, Linux/Unix, CI/CD, MS Office

Certifications: Microsoft Azure Fundamentals (AZ-900), Analytics Vidhya AI/ML Bootcamp